AGILE LAB FAT

QUESTION:SMART ELECTRICITY BILLING SYSTEM

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Smart Electricity Billing System
Problem Statement:

Ensure electricity usage is calculated and billed correctly.
Tasks:

1)Develop billing calculation logic

2)Write JUnit tests for tariff calculations

3)Implement CI/CD pipeline with automated testing

• Set up version control using GitHub

Configure CI pipeline (e.g., Jenkins/GitHub Actions) to trigger on code commitsAutomate build process using MavenIntegrate automated unit tests in pipeline

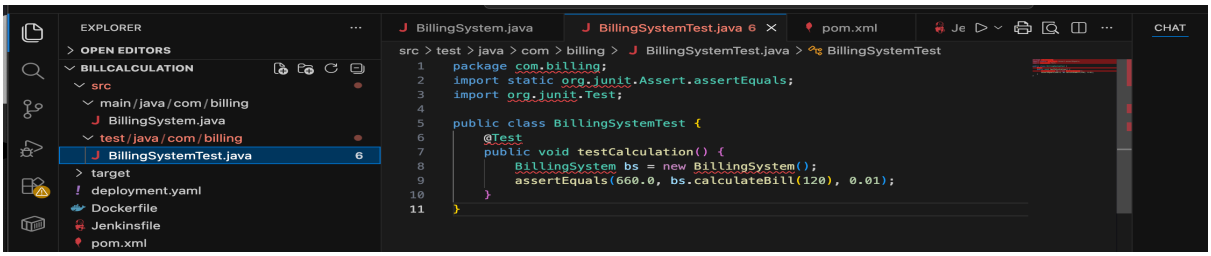
◦ Configure CD pipeline for automatic deployment to staging/production

4) Deploy services on Docker and Kubernetes

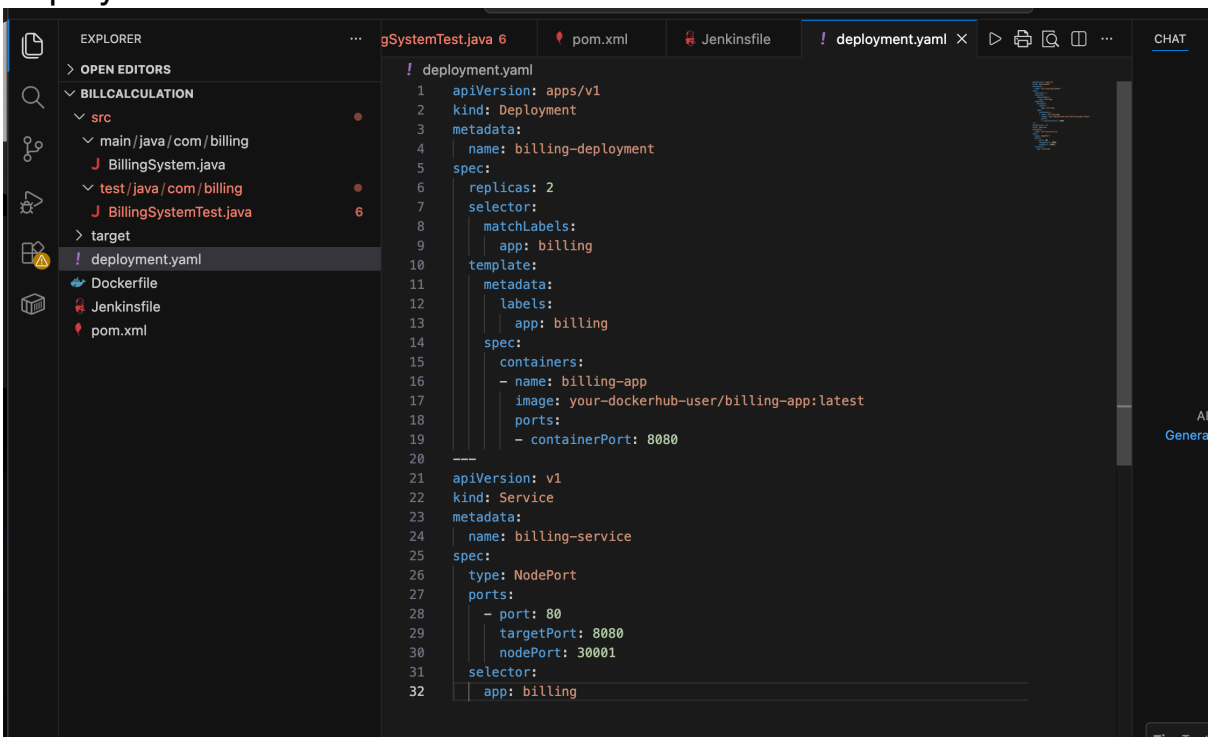
1) Develop billing calculation logic

Code:
Java code:

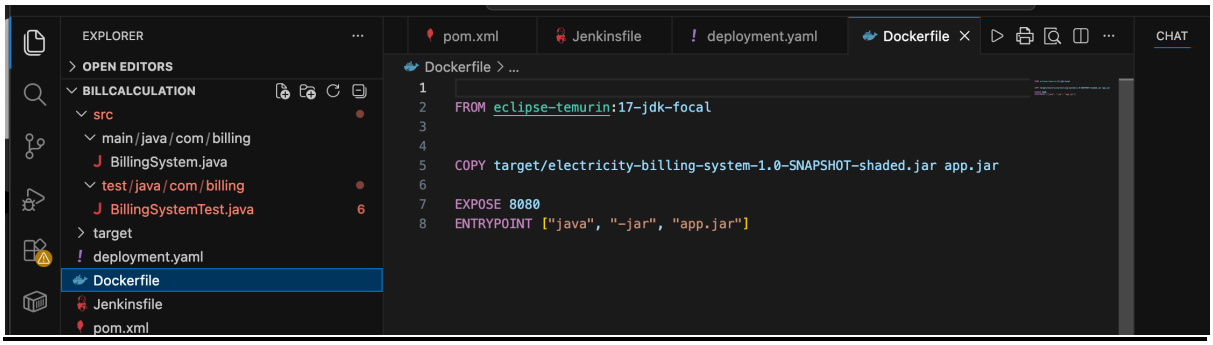
2)Junit test:



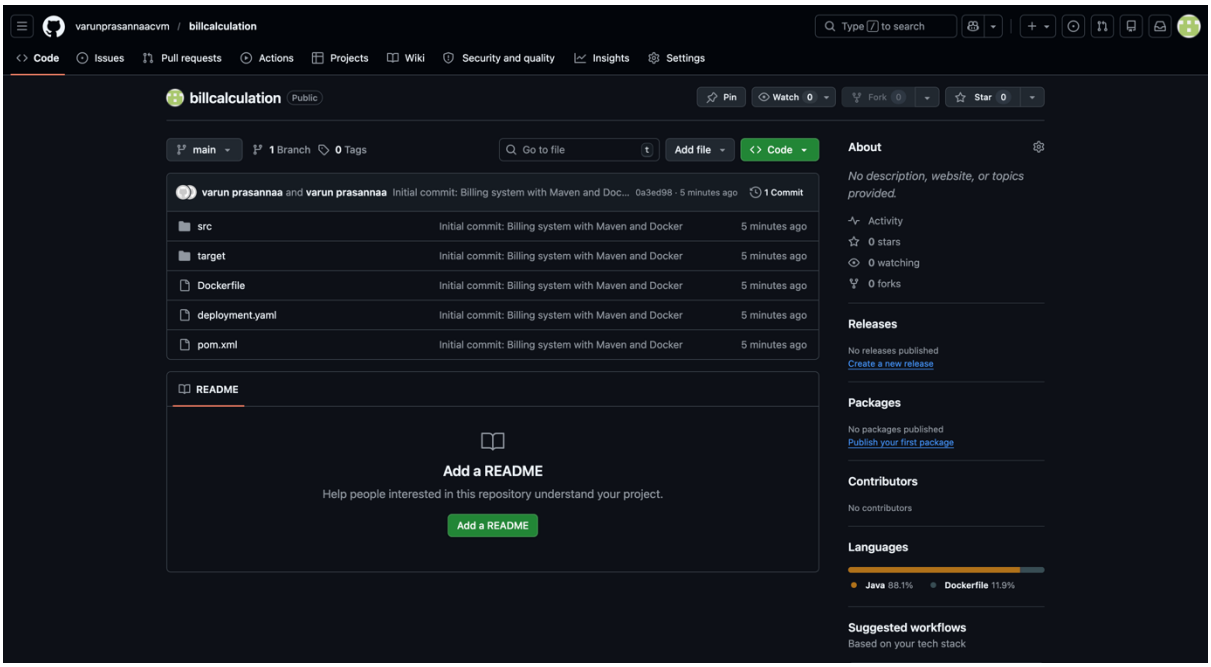
Deployment:



Docker:



3) Implement CI/CD pipeline with automated testing:



Jenkins / BillingSystem-Pipeline / #6

```
#8 unpacking to docker.io/library/billing-app:latest 0.0s done
#8 DONE 0.3s
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Kubernetes Deploy)
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] withEnv
[Pipeline] {
[Pipeline] sh
+ kubectl apply -f deployment.yaml
deployment.apps/billing-deployment created
service/billing-service created
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
Deployment Successful!
[Pipeline] junit
Recording test results
[Checks API] No suitable checks publisher found.
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Build Queue

No builds in the queue.

Build Executor Status 0/2

S	W	Name ↓	Last Success	Last Failure	Last Duration
✓		BillingSystem-Pipeline	14 min #6	16 min #5	58 sec

Pipeline script

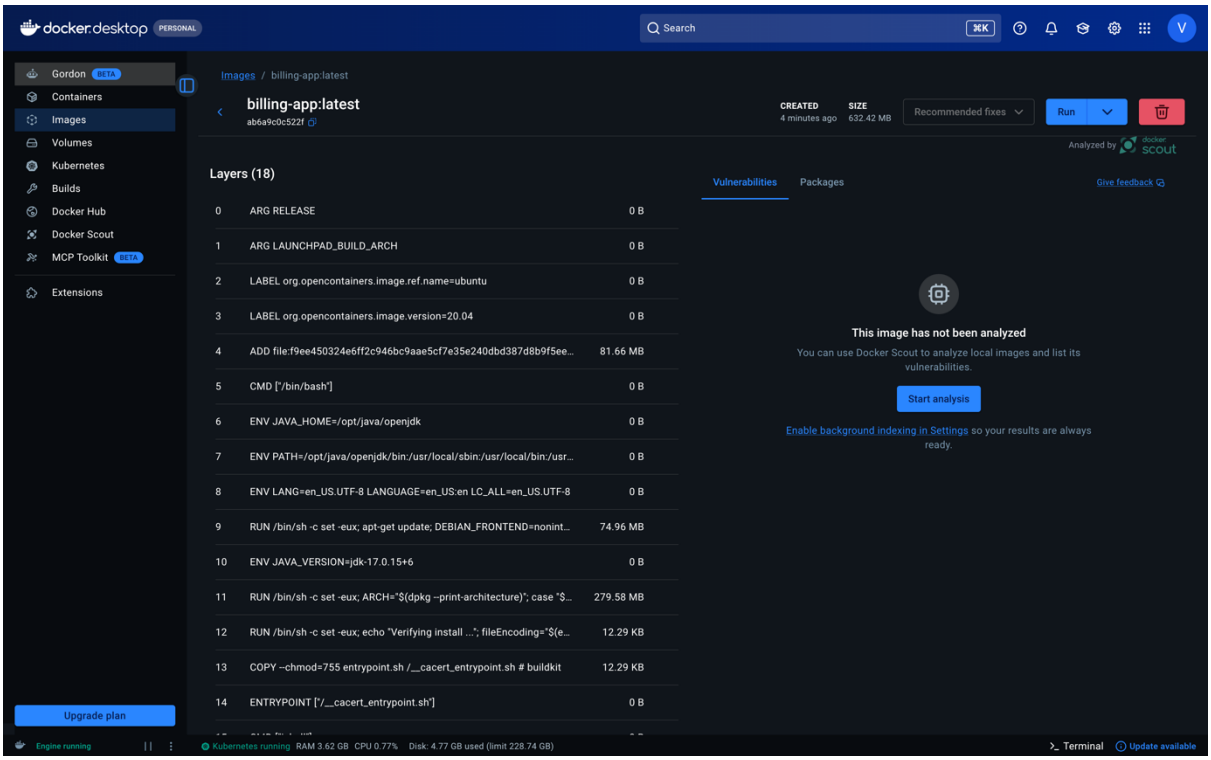
Script ?

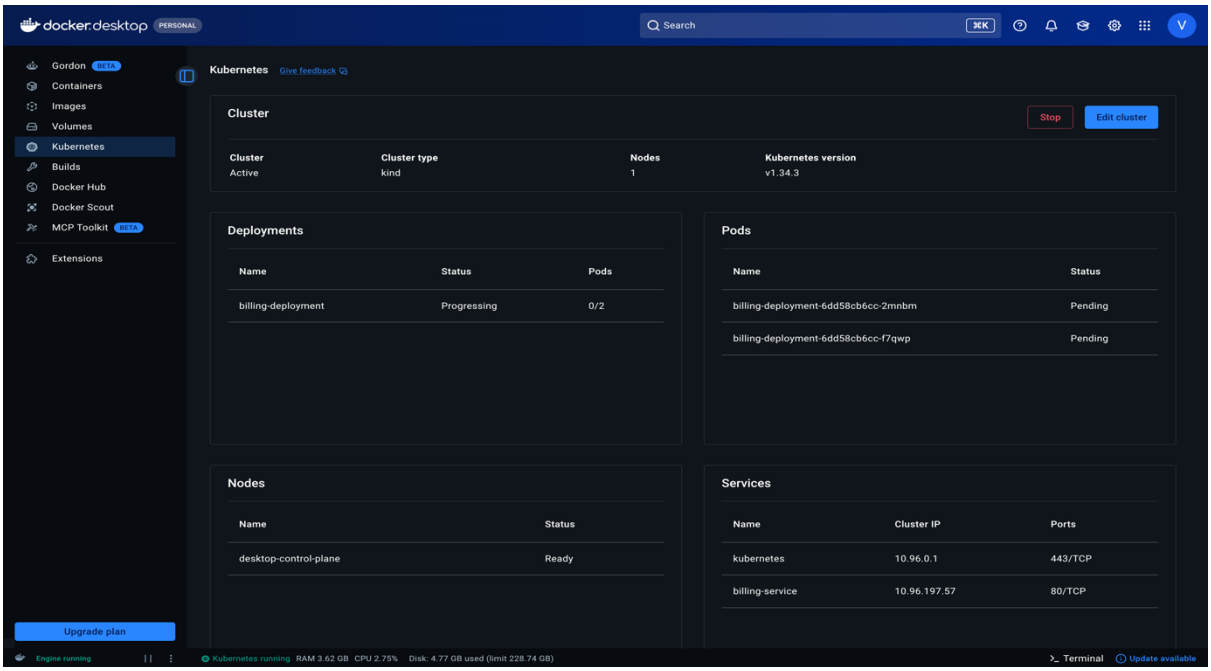
```
pipeline {
    agent any

    tools {
        // Must match the name in Manage Jenkins -> Tools -> Maven
        maven 'Maven3'
    }

    stages {
        stage('Checkout Code') {
            steps {
                // Pulls your code from GitHub
                git branch: 'main', url: 'https://github.com/varunprasannaacvm/billcalculation.git'
            }
        }
```

☑ Use Groovy Sandbox ?

4. Deploy services on Docker and Kubernetes:

Images:


docker desktop PERSONAL
Search ⌘K
Gordon BETA
Containers
Images
Volumes
Kubernetes
Builds
Docker Hub
Docker Scout
MCP Toolkit BETA
Extensions
Upgrade plan
Images / billing-app:latest
billing-app:latest
ab6a9c0c522f
CREATED 4 minutes ago
SIZE 632.42 MB
Recommended fixes
Run
Analyzed by docker scout
Layers (18)
0 ARG RELEASE 0 B
1 ARG LAUNCHPAD_BUILD_ARCH 0 B
2 LABEL org.opencontainers.image.ref.name=ubuntu 0 B
3 LABEL org.opencontainers.image.version=20.04 0 B
4 ADD file:f9ee450324e6ff2c946bc9aae5cf7e35e240dbd387d8b9f5ee... 81.66 MB
5 CMD ["/bin/bash"] 0 B
6 ENV JAVA_HOME=/opt/java/openjdk 0 B
7 ENV PATH=/opt/java/openjdk/bin:/usr/local/sbin:/usr/local/bin:/usr... 0 B
8 ENV LANG=en_US.UTF-8 LANGUAGE=en_US:en LC_ALL=en_US.UTF-8 0 B
9 RUN /bin/sh -c set -eux; apt-get update; DEBIAN_FRONTEND=nonint... 74.96 MB
10 ENV JAVA_VERSION=jdk-17.0.15+6 0 B
11 RUN /bin/sh -c set -eux; ARCH="$(dpkg --print-architecture)"; case "$... 279.58 MB
12 RUN /bin/sh -c set -eux; echo "Verifying install ...'; fileEncoding="$(e... 12.29 KB
13 COPY --chmod=755 entrypoint.sh /__cacert_entrypoint.sh # buildkit 12.29 KB
14 ENTRYPOINT ["/__cacert_entrypoint.sh"] 0 B
Vulnerabilities
Packages
Give feedback
This image has not been analyzed
You can use Docker Scout to analyze local images and list its vulnerabilities.
Start analysis
Enable background indexing in Settings so your results are always ready.
Engine running
Kubernetes running RAM 3.62 GB CPU 0.77% Disk: 4.77 GB used (limit 228.74 GB)
Terminal
Update available


docker desktop PERSONAL
Search ⌘K
Gordon BETA
Containers
Images
Volumes
Kubernetes
Builds
Docker Hub
Docker Scout
MCP Toolkit BETA
Extensions
Upgrade plan
Kubernetes Give feedback
Cluster
Stop
Edit cluster
Cluster Active
Cluster type kind
Nodes 1
Kubernetes version v1.34.3
Deployments
Name Status Pods
billing-deployment Progressing 0/2
Pods
Name Status
billing-deployment-6dd58cb6cc-2mnbm Pending
billing-deployment-6dd58cb6cc-f7qwp Pending
Nodes
Name Status
desktop-control-plane Ready
Services
Name Cluster IP Ports
kubernetes 10.96.0.1 443/TCP
billing-service 10.96.197.57 80/TCP
Engine running
Kubernetes running RAM 3.62 GB CPU 2.75% Disk: 4.77 GB used (limit 228.74 GB)
Terminal
Update available

```
Last login: Wed Apr 15 16:31:22 on ttys000
varunprasannaa@varuns-MacBook-Pro ~ % kubectl get pods
No resources found in default namespace.
varunprasannaa@varuns-MacBook-Pro ~ % kubectl get service billing-service
Error from server (NotFound): services "billing-service" not found
varunprasannaa@varuns-MacBook-Pro ~ % kubectl get pods
NAME                                  READY   STATUS         RESTARTS   AGE
billing-deployment-6dd58cb6cc-2mnbm   0/1     ErrImagePull   0          2m2s
billing-deployment-6dd58cb6cc-f7qwp   0/1     ErrImagePull   0          2m2s
varunprasannaa@varuns-MacBook-Pro ~ % kubectl get service billing-service
NAME              TYPE       CLUSTER-IP     EXTERNAL-IP   PORT(S)        AGE
billing-service   NodePort   10.96.197.57   <none>        80:30001/TCP   2m9s
varunprasannaa@varuns-MacBook-Pro ~ %
```